Department of Mechanical, Aerospace and Industrial Engineering
Ryerson University
Mid-Term Test 1-Winter 2002
Materials Science I (MTL 200)

Test Duration: 60 mins

Name (Last-First): [redacted] , [redacted] **Section No:** 13
ID: [redacted]

Instructions:
1. This is a closed book test.
2. You may use an electronic calculator.
3. For **multiple choice questions** choose one and only the best answer.
4. For numerical problems, the required equations are given. **No unit conversion is required.**
5. Back of pages can be used if extra pages are needed.

Marks	
MC	14/15
10	3/10
11	14/25
12	22/25
13	15/25
Bonus	5/5
Total	76

81

Multiple-choice Questions

1. Burger's vector presents: **[1-Mark]**
   (a) The magnitude of material deformation.
   (b) Both the magnitude and the direction of material deformation.
   (c) The direction of dislocation motion.
   (d) The lattice distance in a crystal structure.

2. When hydrogen electron being excited into a lower orbit, **[1-Mark]**
   (a) Energy is absorbed.
   (b) Energy of a photon becomes unstable.
   (c) Energy is emitted.
   (d) Electromagnetic radiation is produced.

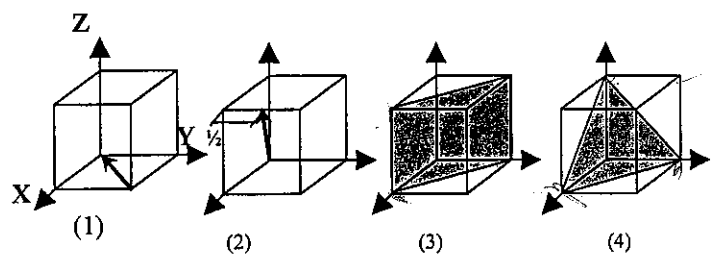
3. **Cl** atom and **Na** atom have atomic numbers of 17 and 11 respectively. In an NaCl molecule, **[1-Mark]**
   (a) **Cl** atom gives up valence electrons and acts as electropositive.
   (b) **Na** atom attracts valence electrons and acts as electropositive.
   (c) **Cl** atom attracts valence electrons and acts as electronegative
   (d) **Na** atom gives up valence electrons and acts as electronegative.

4. Which of the following answers presents atomic bonding for MgO, H_2O and Fe, respectively. **[1-Mark]**
   (a) Ionic, Covalent, Metallic
   (b) Metallic, Ionic, Covalent

(c) Ionic , Hydrogen, Metallic
(d) Covalent, Metallic, Ionic

5. In solid solution, there are at least two types of elements involved: solvent and solute elements. In sterling silver composed of 92.5% silver and 7.5% copper: **[1-Mark]**
   (a) Silver atoms represent solvent element, while Copper atoms represent solute element.
   (b) Copper atoms represent solvent element, while Silver atoms represent solute element.

6. For Crystallographic directions and planes shown below which of the following answers is correct. **[3-Marks]**
   (a) 1: $[\bar{1}\bar{1}0]$, 2: (210), 3: $(\bar{1}01)$, 4: [111]
   (b) 1: $(\bar{1}\bar{1}0)$, 2: (212), 3: (101), 4: (111)
   (c) 1: $[\bar{1}\bar{1}0]$, 2: [212], 3: (110), 4: (111)
   (d) 1: $[\bar{1}\bar{1}0]$, 2: [121], 3: (110), 4: (111)



7. Factors affecting diffusion and activation energy are: **[1-Mark]**
   (a) Temperature, melting point and atomic packing factor
   (b) Atomic weight, crystal structure and valences
   (c) Temperature, Atomic weight and electronegativity
   (d) Atomic size, Materials bonding and crystal structure

8. What is the ASTM grain size number of a metal if there are N=256 grains/in^2. (use $N=2^{n-1}$). **[3-Marks]**
   (a) n=9
   (b) n=7
   (c) n=8
   (d) n=6.5

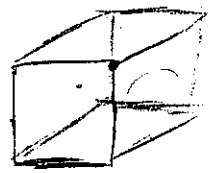
$\ln N = n-1(\ln 2)$
$\frac{\ln 256}{\ln 2} = n-1$
$n = \frac{\ln 256}{\ln 2} + 1$

9. Number of atomic sites per cubic meter for copper is equal to $N=8\times10^{28}$, which of the following answers corresponds to number of vacancies at T=1273 K? Activation energy Q_d=0.9 eV and gas constant $k=8.62\times10^{-5}$ eV/K. **[3-Marks]**
   (a) $N_v=2.2\times10^{25}$ vacancies/m^3
   (b) $N_v=9.2\times10^{25}$ vacancies/m^3
   (c) $N_v=1.2\times10^{25}$ vacancies/m^3
   (d) $N_v=8.0\times10^{25}$ vacancies/m^3

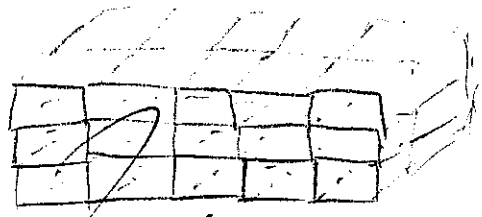
$$N_v = N\exp\left(\frac{-Q_d}{kT}\right)$$

Questions

10. Explain how ductility of polycrystalline materials is different from single crystal materials. Enclose your answer with sketches of the microstructure of polycrystalline and single crystal materials. Give an example of each material. **[10-Marks]**



Single crystal
(NaCl)



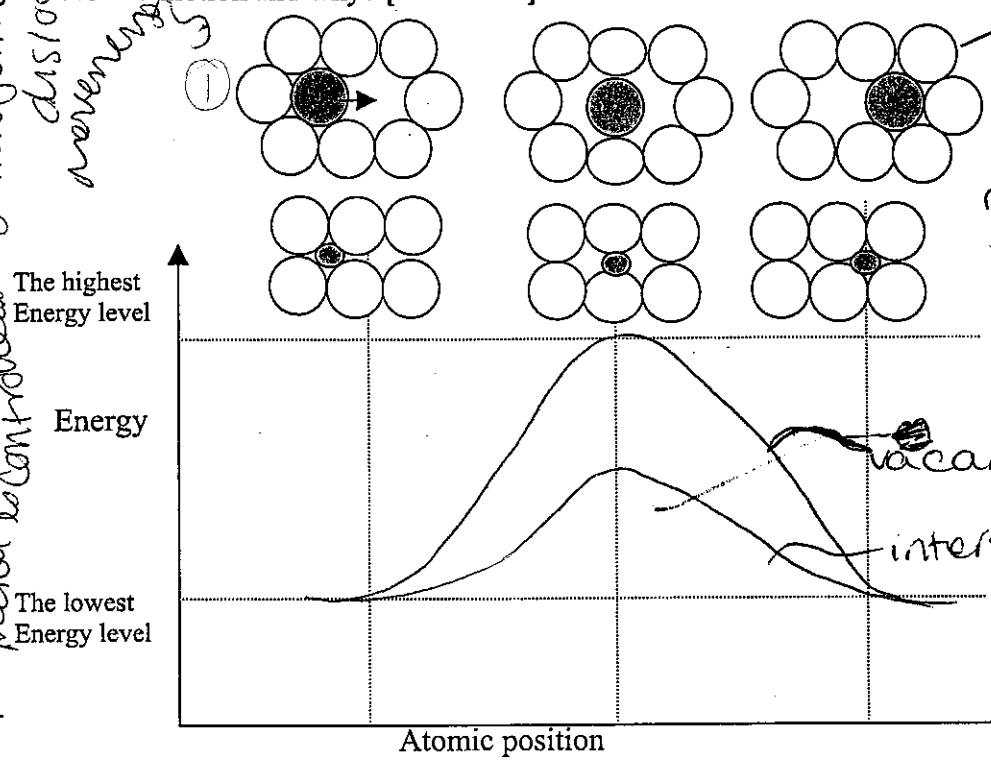
Polycrystalline

3/10

Polycrystalline materials tend to be more ductile than single-crystal materials. This is because there are more areas in the polycrystalline structure bend with a given force. The force is spread out among the entire structure. whereas, with the single crystal structure, there isn't very much ductility because the force is concentrated almost to a single point, therefore there is no room to bend

14/25

11. (a) Schematically show how activation energy changes as a diffusing atom moves from one site to an adjacent site for both diffusion mechanisms of vacancy and interstitial. Which mechanism may require to possess a greater activation energy for atomic motion and why? **[10-Marks]**



vacancy requires a greater activation energy because the force required to squeez through the other atoms is greater due to the increase size of the atom.

atomic size?!

force required for slip less than 1/100 force to break metal bonds

slip provides ductility & peak properties of metal is controlled by interfering w/ dislocation movements

(b) (i) Briefly explain significance of dislocations in materials deformation.
(ii) After Rolling of a metallic plate, its thickness reduces, and grains on the plate thickness reduce in size. Briefly explain how changes in grain size can affect the strength of this metallic plate.
(iii) Name important parameters required to be satisfied for a solid solution.
[15-Marks]

i) Dislocations in materials increase the strength and ductility of the material. There are many different types of deformations, such as interstitial, Schottky and frankel

2/5

ii) Generally, the smaller the grain size is, the more grain boundaries there are, this results in a stronger, more ductile metallic plate. Therefore, the plate that has been rolled is now stronger than it was before

4/5

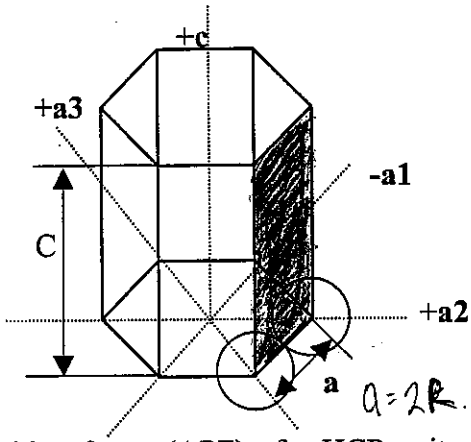
iii)

0/5

Numerical Problems

$\frac{22}{25}$

12. (a) Draw crystallographic plane $(0\ 1\bar{1}\ 0)$ in the HCP unit-cell shown.

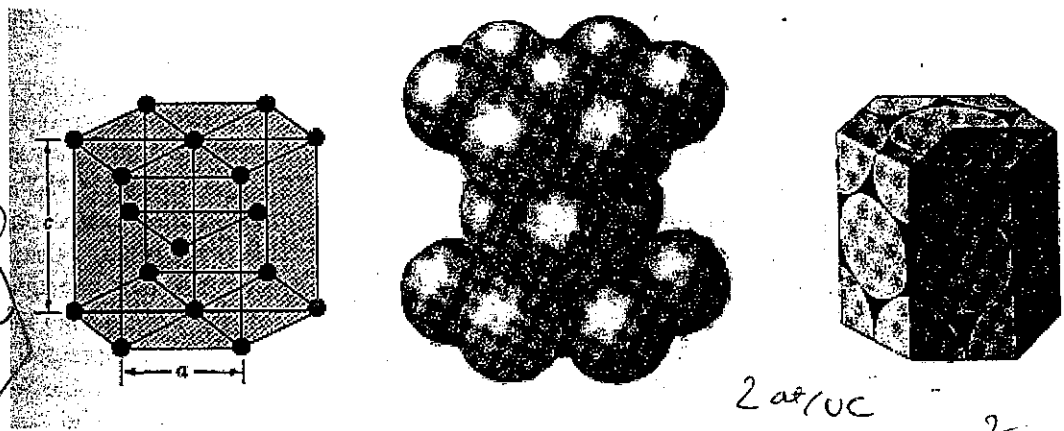


$a = 2R$

$\frac{10}{10}$

(b) Show the atomic packing factor (APF) of a HCP unit-cell is equal to 0.74 if the lattice parameter ratio C/a is 1.633 (C is the height of HCP unit-cell). **[25-Marks]**

$\frac{2\pi}{3\sqrt{3}(1.633)}$



$a = 2R$

$c = 2a(1.633)$

$V_c = \frac{3\sqrt{3}a^2c}{2}$

2 at/UC

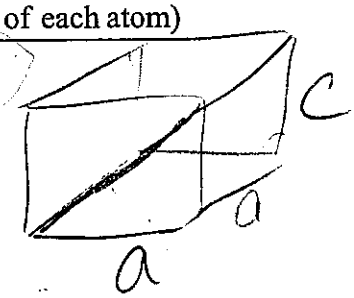
$\frac{2}{3}\pi R^3$

$\frac{\pi}{3\sqrt{3}(1.633)}$

$\frac{6 \times \frac{4}{3}\pi R^3}{3\sqrt{3}a^2 \times 1.633}$

$\frac{\pi}{3\sqrt{3}(1.633)}$

$$APF = \frac{\text{(No. of atoms/unit - cell)} \times \text{(Volume of each atom)}}{\text{Volume of unit - cell}}$$



$\frac{C}{a} = 1.633$

$a = \frac{C}{1.633}$

$$\text{Volume of each atom} = \frac{4}{3}\pi R^3$$

$\frac{a}{2} = R$

2at/UC = 3 UC/HCP

$\therefore V_{UC} = 6\left(\frac{4}{3}\pi R^3\right)$

$= 8\pi\left(\frac{a^3}{8}\right)$

$\frac{12}{15}$

$\frac{C}{2R} = 1.633$

$C = 3.266R$

$V_{UC} = a^3\pi$

$V_{at} = 0.5235\,a^3$

$V = a \times a \times C$

$= \frac{3.266R}{1.633} \times \frac{3.266R}{1.633} \times 3.266R$

$= 2R \times 2R \times 3.266R$

$V_c = 13.064R^3$

PACK

13. Determine the time it takes to obtain a carbon concentration of 0.24% at a depth of 0.01 cm beneath the surface of an iron bar when carburizing at 1273 K (*Carburizing* is a process by which the surface carbon concentration of a ferrous alloy is increased by diffusion from the surrounding environment). The initial concentration of carbon in the iron bar is 0.20% and the surface concentration in a CO/CO_2 gas environment is maintained at 0.40%. In this problem, the Fe has the FCC structure and the diffusion coefficient is D= $(2\times10^{-5}$ m^2/sec){exp[(-142,000 J/mol)/kT), where gas constant k=8.314 J/mol-K. **[25-Marks]**

Use the Fick's second law and error function table given:

$x = 0.01$ cm $= 0.0001$ m

$T = 1273$

$C_0 = .20$

$C_s = .40$

$C_x = 0.24$

$$(C_s - C_x)/(C_s - C_0) = erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

z	0.70	0.75	0.80	0.85	0.90	0.95	1.0
erf (z)	0.6778	0.7112	0.7421	0.7707	0.7970	0.821	0.843

$$\frac{C_s - C_x}{C_s - C_0} = erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$D = (2\times10^{-5}) \exp\left(\frac{-142000}{8.314 \times 273}\right)$$

$$= 2.98 \times 10^{-11}$$

$$\frac{0.40 - 0.24}{0.40 - 0.20} = erf\left(\frac{0.0001}{2\sqrt{(2.98\times10^{-11})(t)}}\right)$$

$$\frac{0.16}{0.20} = erf\left(\frac{0.0001}{2\sqrt{2.98\times10^{-11}(t)}}\right)$$

$$0.8 = erf\left(\frac{0.0001}{2\sqrt{2.98\times10^{-11}(t)}}\right)$$

See the table carefully!

$$\therefore \frac{0.0001}{2\sqrt{2.98\times10^{-11}(t)}} = 0.7421$$

$erf(z) = 0.8$

$z \approx 0.90$

$$\left(\frac{0.0001}{2}\right)^2 = 0.7421(2.98\times10^{-11})(t)$$

$$\frac{2.0\times10^{-9}}{(0.7421)(2.98\times10^{-11})} = t$$

$113 = t$

$t = 113$ s

18/25

BONUS Question

14. (a) Calculate the spacing of planes responsible for a diffraction angle of 64.4°, assuming first-order diffraction. The wavelength of the radiation is 1.54A. (b) Which one of the following crystallographic planes corresponds to the atomic lattice distance of a=2.502A. **[5-Marks]**

Planes	(110)	(111)	(210)

$2d\sin\theta = n\lambda$

$$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$$

$\theta = 32.2°$

~~$a = 2.502 \times 10^{-10}$~~

~~$\lambda = 1.54 \times 10^{-10}$~~

$n = 1$

a) $2d\sin\theta = n\lambda$

$d = \frac{n\lambda}{2\sin\theta} = \frac{(1)(1.54A)}{2\sin(32.2)} = 1.44A.$ ✓

$\frac{5}{5}$

b) $1.44A = \frac{2.502A}{\sqrt{h^2+k^2+l^2}}$

$\sqrt{h^2+k^2+l^2} = \frac{2.502A}{1.44A} = 1.7375$

$(110) = \sqrt{1^2+1^2+0} = \sqrt{2} \neq 1.414$

$(111) = \sqrt{1^2+1^2+1^2} = \sqrt{3} = 1.73$

$(210) = \sqrt{2^2+1^2+0^2} = \sqrt{5} = 2.236$

$\therefore$ (111) corresponds to the lattice distance of a=2.502A.